

alpaca

Autonomous options trading with deterministic risk control

Gemini reads the market and selects an approved structure. Code controls every leg, size, limit, and exit.



PAPER TRADING

Nine ETFs across three independently managed paper accounts

SEP 2026

Models can read markets. Limits need code.

A useful trading agent must reason under uncertainty without gaining authority over the account's safety boundaries.

AI IS GOOD AT

Reading regime and volatility

Interpreting news and catalysts

Comparing strategy tradeoffs

CODE MUST OWN

Position sizing and maximum loss

Liquidity and event constraints

Orders, exits, and kill switches

Alphaca separates judgment from authority.

Six roles turn market data into a managed position

- | | | | |
|----|---|---------------------|---|
| 01 | — | Scout | Measures trend, realized volatility, implied volatility, and their spread |
| 02 | — | Analyst | Uses Gemini with search grounding to capture news and scheduled catalysts |
| 03 | — | Strategist | Compares approved structures and returns a proposal reference |
| 04 | — | Risk officer | Applies deterministic gates and computes a safe quantity |
| 05 | — | Executor | Places reproducible multi-leg orders through the Alpaca CLI |
| 06 | — | Manager | Monitors fills, profit targets, stops, and expiration safety |

The strategist chooses from trades that already passed risk review

The model never emits a strike, quantity, price, or option leg.

1. CANDIDATE FACTORY

Deterministic code builds defined-risk structures from the live chain.

2. RISK VERDICT

Each proposal receives a named approval or rejection before the model sees it.

3. SELECTION

Gemini returns an approved proposal ID or declines to trade.

A confused model can choose a worse safe trade. It cannot create an unsafe one.

Eight checks stand between an idea and an order

Current barbell preset. Limits are explicit, tested, and journalled by name.

G1 **Kill switches**
4% day loss, 10% total drawdown

G2 **Trade budget**
7 new trades, 10 open structures

G3 **Defined risk**
Every short leg requires matching cover

G4 **Liquidity**
Rejects wide or unusable quotes

G5 **Pricing**
18% credit floor, 45% debit cap

G6 **Event blackout**
Blocks premium selling across catalysts

G7 **Position sizing**
1% tactical, 1.25% carry per trade

G8 **Portfolio limits**
8% open risk with sleeve and symbol caps

Naked short options are structurally impossible. No configuration flag disables the coverage rule.

Three accounts test different sources of return on the same tape

BARBELL

Balanced

**Core income, convex trades,
and financed carry**

Best for changing volatility regimes

CONVEX TILT

Asymmetric

**More budget for debit
spreads and long gamma**

Benefits from large directional moves

INCOME ONLY

Control

**Defined-risk premium selling
without convexity**

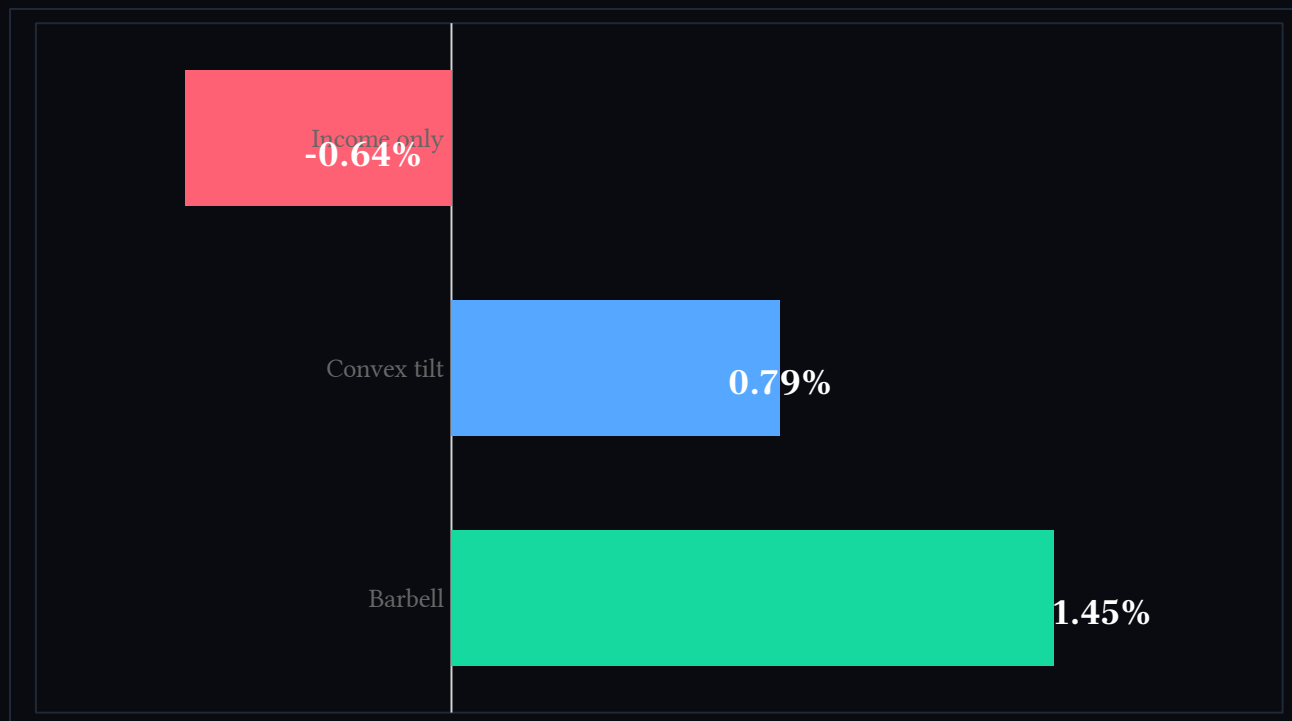
Tests whether volatility risk premium alone works

Strategy set: credit spreads, debit spreads, iron condors, and financed risk reversals

Universe: SPY, QQQ, IWM, DIA, XLK, XLF, SMH, GLD, and TLT

The balanced account leads the live three-account test

Broker equity return as shown on 4 September 2026. The sample covers fewer than five sessions.



MAIN ACCOUNT EQUITY

\$101,451.99

+\$1,451.99 from \$100,000

PEAK EQUITY

\$101,965.56

+1.97% intraday peak

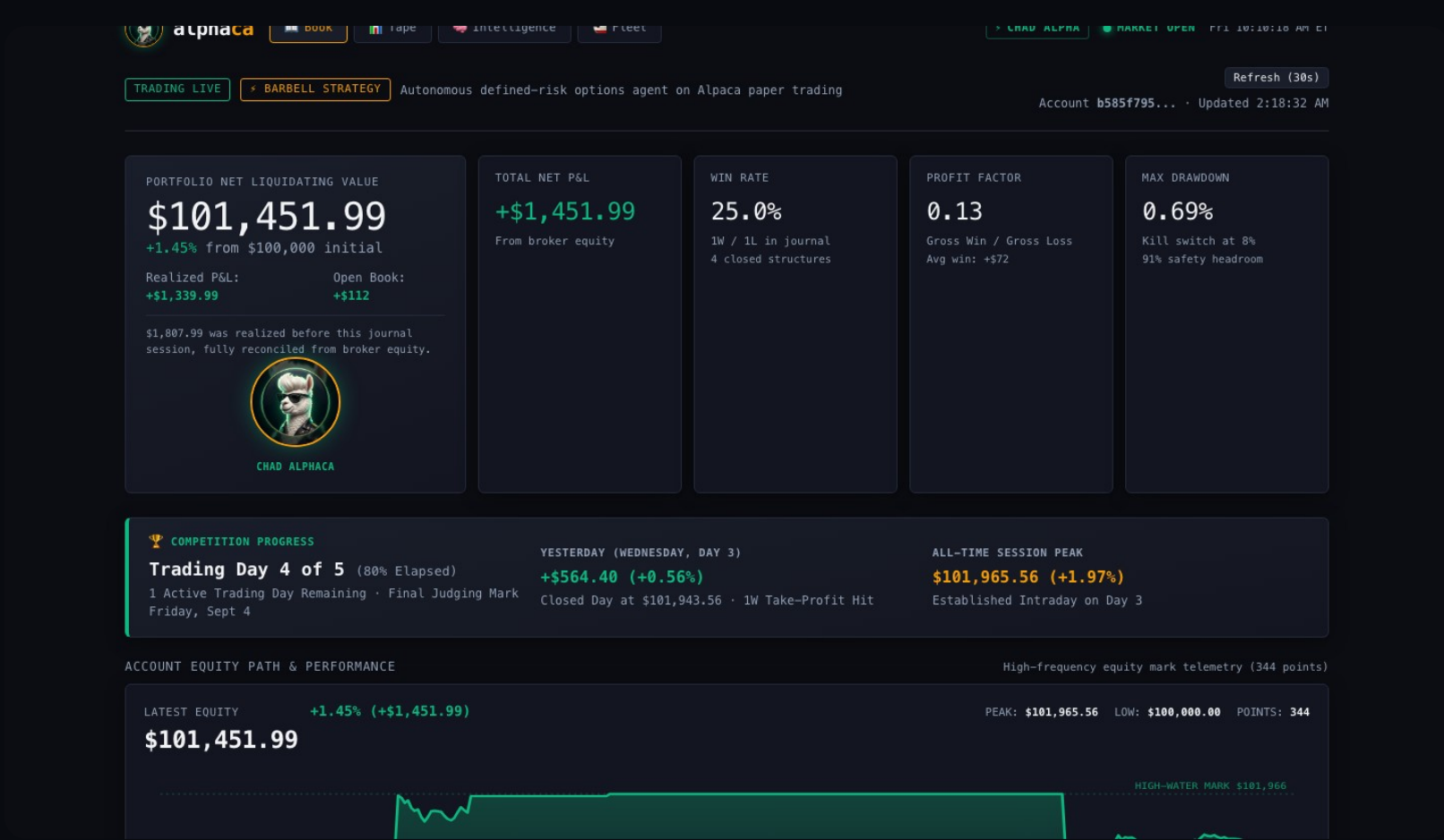
MAXIMUM DRAWDOWN

0.69%

Observed on the main account

Paper results are experimental. They do not establish a durable edge.

Every fill, refusal, and risk decision is visible



PUBLIC DASHBOARD

Live broker equity

Portfolio Greeks and payoff curves

Open and closed structures

Named rejection reasons

Stress tests and session plans

Public link on the closing slide

Autonomy without unchecked financial authority

Alphaca shows how an AI trading agent can act independently while deterministic software keeps control of account risk.

MARKET COVERAGE

9 ETFs

Liquid equity, sector, commodity, and bond proxies

PORTFOLIO EXPERIMENT

3 accounts

Balanced, convex tilt, and income control

OPEN RISK CEILING

8%

Defined risk with a 10% drawdown stand-down

Repository

github.com/TheSupermanish/superio-trading-agent

Live demo

thesupermanish.github.io/superio-trading-agent

Built on Alpaca paper trading with Gemini, Vertex AI, Python, and Next.js